

Mert Ali Türk

Embedded Systems Engineer

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🔗 Portfolio Website 📺 Youtube

PROFILE

Master's student in Embedded Systems Design with a focus on embedded software and HW/SW co-design.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer Intern

04/2026

ZARM Technik AG

Bremen, Germany

- Assisted in the development and maintenance of the Magnetic Attitude Driving System ECU software for Release 2.0.0.
- Fixed software bugs and performed software refactoring to improve code quality.
- Developed PID control algorithm for magnetic torquers and ADC peripheral driver for ATmega microcontroller.
- Applied MISRA-C/C++ guidelines using CppCheck and Clang for static analysis
- Used Git/GitLab for version control and CI/CD workflows

Electrical Engineer Intern

07/2024 – 09/2024

ESENYURT BELEDIYESI

Turkey

- Gained hands-on experience in technical documentation, systematic problem-solving, and adherence to industry standards (ISO/IEC 27002:2022) in a technical environment.

EDUCATION

M.Sc. in Embedded Systems Design (GPA: 1.26, German scale)

04/2025 – Present

Hochschule Bremerhaven

Germany

- **Relevant Courses:** Digital Design, C/C++ , VHDL, SoC ,Model-Based /Real-Time Software Development, Requirement Engineering, ARP4761

B.Sc. Electrical and Electronic Engineering (GPA : 3.13)

09/2018 – 06/2024

Zonguldak Bulent Ecevit University

Turkey

- Thesis: “Road-Embedded Solar Panel Integrated Wireless Electric Vehicle Charging System” —Designed and validated an embedded IoT energy management system using a microcontroller for real-time monitoring and control.

Telecommunication Engineering (Erasmus+ Exchange Student.)

01/2021 – 06/2021

Universidade Vigo

Spain

- Specialized in telecommunications, computer networks, C/C++

PROJECTS

Measurement Network Gateway

- Designed an embedded HW/SW co-designed networking gateway on a Zynq to collect and forward sensor data. Developed FPGA logic to sample sensors and used PetaLinux for networking. Enabled reliable real-time data transmission to a GUI interface.

Skills: *HW/SW Co-design, Petalinux, Zynq, VHDL, Embedded Linux*

8-Bit Microprocessor Design

- Designed and built a functional 8-bit microprocessor using 74xx series logic chips to execute instructions and demonstrate complete control and data flow.

Skills: *Computer Architecture, Datasheet Reading*

SKILLS

Programming Languages: — VHDL, C/C++, Python

Protocols & Standards: — Ethernet , CAN , UART , I2C , SPI , DMA , MISRA-C/C++ , AUTOSAR, DO-178B/C, ARINC 653, ARP4761, ARP4754

Development Tools: — AMD Vivado, AMD Vitis, STM32CubeIDE, KEIL uVision, MATLAB/Simulink, IBM Rhapsody, Git, Make/Makefile, CMake, Docker, CI/CD, IBM DOORS, Isograph Reliability Workbench

Embedded Boards: — Arty Z7 ,ZedBoard ,STM32F4/7 ,ESP32

Operating Systems & Frameworks — Embedded Linux, Yocto Project, Petalinux, FreeRTOS, uC/OS-ii

Testing Methodologies: — Software-in-the-Loop (SIL) Testing, MIL Testing w/Simulink, Embedded Software Verification & Validation, CppUnitTest

AWARDS

Finalist: in der CAMPUSiDEEN 2025 [!\[\]\(f1c5da15572e3e09d343161be98f508d_img.jpg\)](#)

12/11/2000

BRIDGE gründen aus Bremer Hochschulen

- EcoOptima is an initiative focused on energy optimization and has successfully advanced to the finals among more than 100 ideas.

LANGUAGES

English — Fluent

German — Conversational

TOEFL - B2 Certificate

ORGANIZATIONS

Ateşböceği

Core Member Of Ateşböceği Organization

- Volunteered for Ateşböceği Derneği, a non-profit organization supporting children and animals in need.